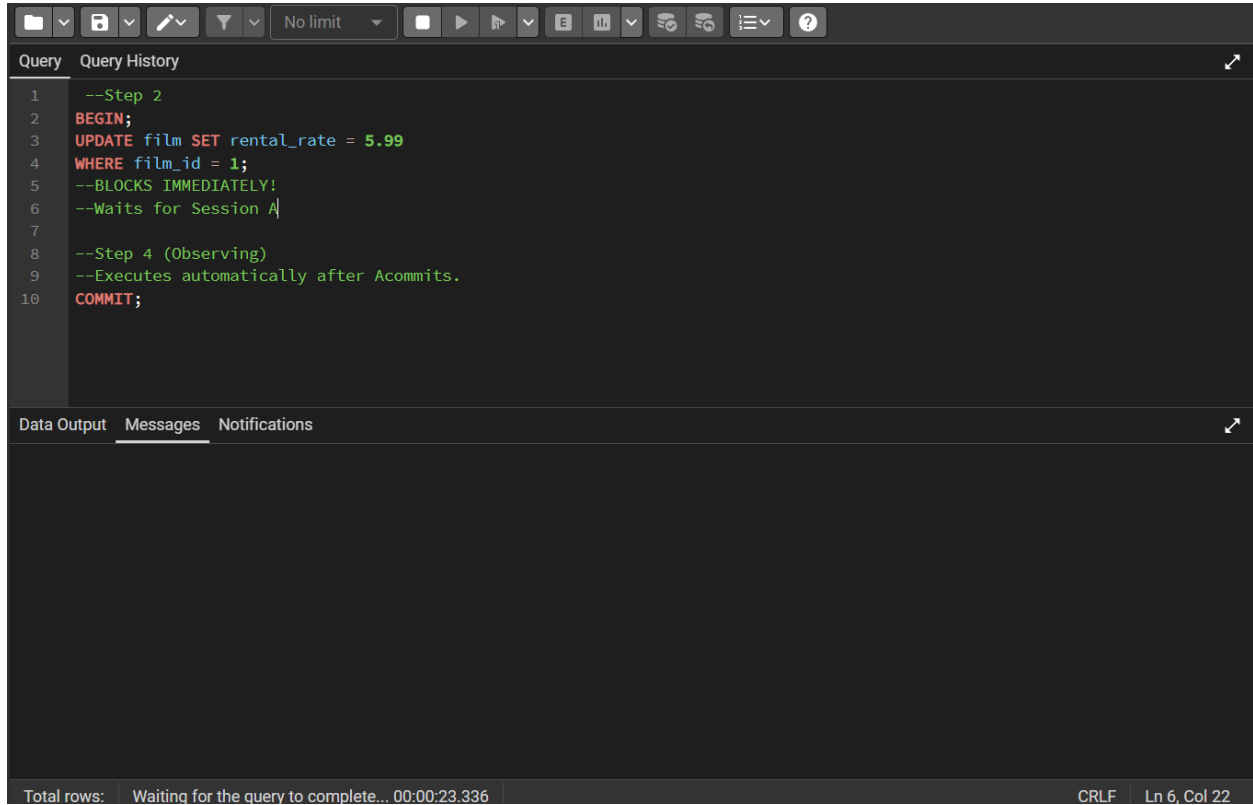


Lab 1 :

Sau khi chạy step 2 session B bị Block không trả về kết quả :

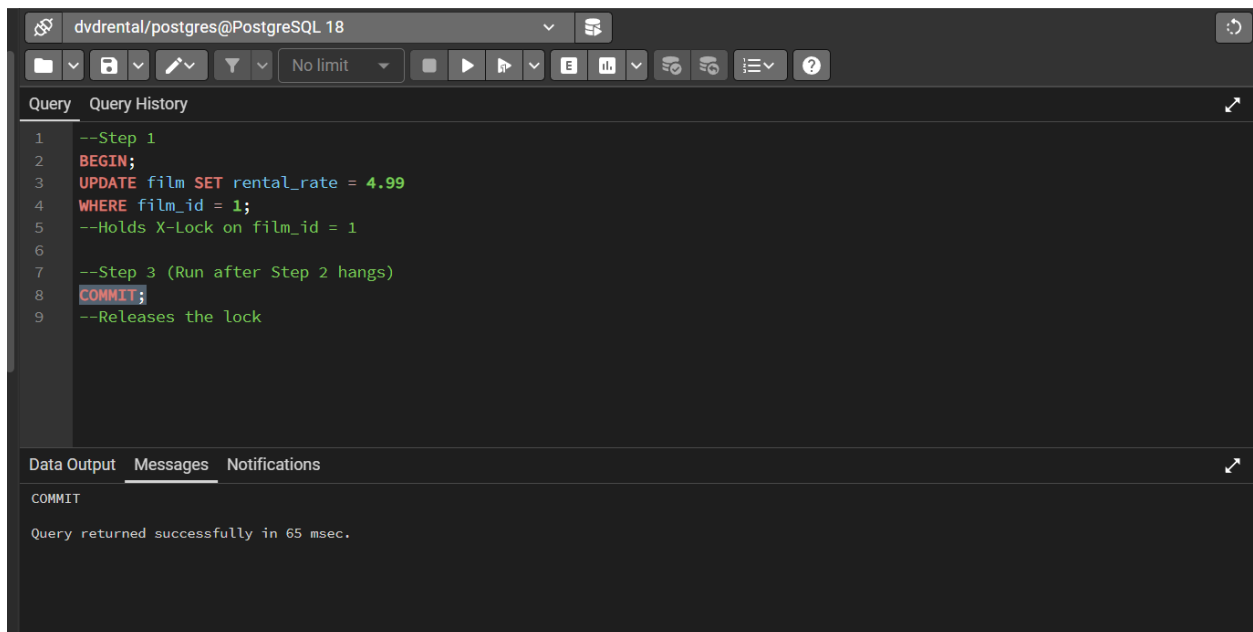


The screenshot shows a PostgreSQL query editor with a dark theme. The query is as follows:

```
1  --Step 2
2  BEGIN;
3  UPDATE film SET rental_rate = 5.99
4  WHERE film_id = 1;
5  --BLOCKS IMMEDIATELY!
6  --Waits for Session A|
7
8  --Step 4 (Observing)
9  --Executes automatically after Acommits.
10 COMMIT;
```

The editor has tabs for "Query", "Query History", "Data Output", "Messages", and "Notifications". The "Messages" tab is active, but it is empty. The status bar at the bottom indicates "Total rows: Waiting for the query to complete... 00:00:23.336" and "CRLF Ln 6, Col 22".

Commit :



The screenshot shows the same PostgreSQL query editor with a different query:

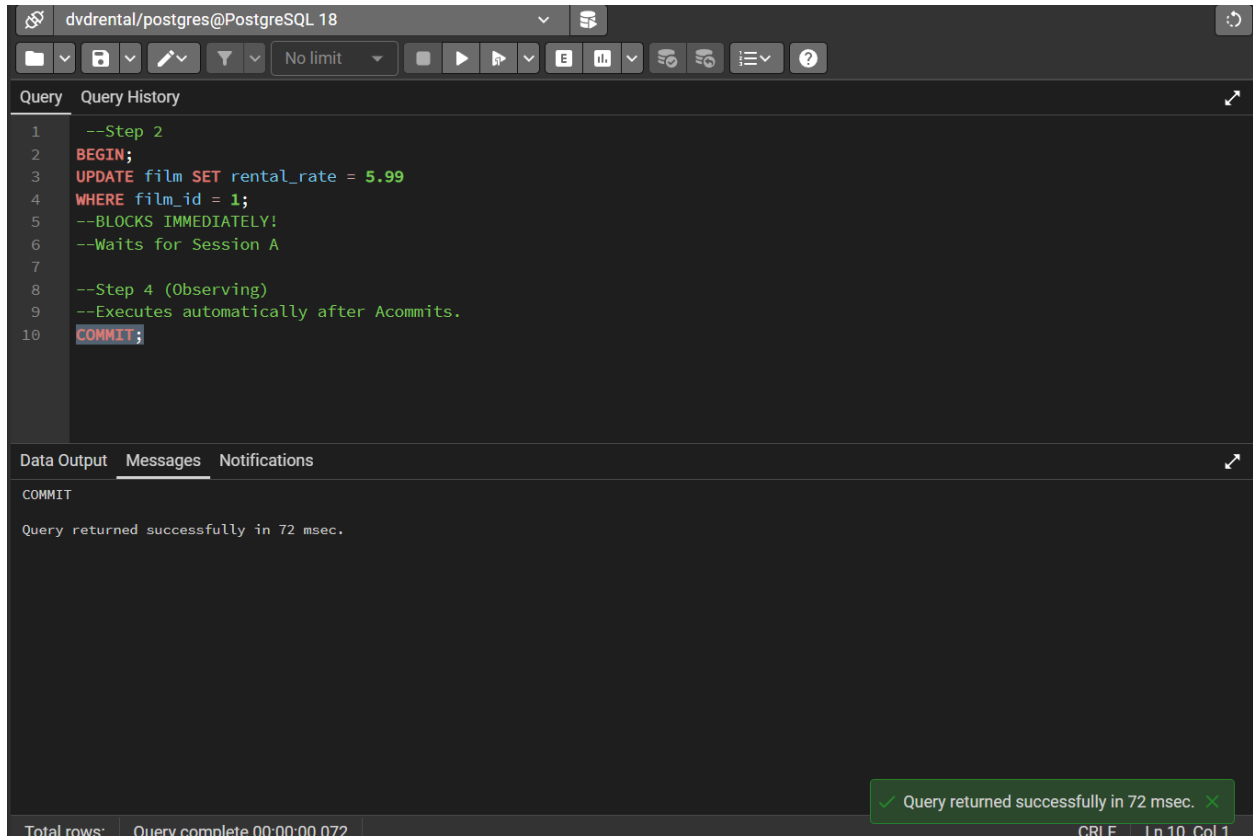
```
1  --Step 1
2  BEGIN;
3  UPDATE film SET rental_rate = 4.99
4  WHERE film_id = 1;
5  --Holds X-Lock on film_id = 1
6
7  --Step 3 (Run after Step 2 hangs)
8  COMMIT;
9  --Releases the lock
```

The "Messages" tab is active and shows the following output:

```
COMMIT
Query returned successfully in 65 msec.
```

The status bar at the bottom is not visible in this screenshot.

Ngay sau khi Session A COMMIT, Session B sẽ tự động chạy tiếp và hoàn thành:

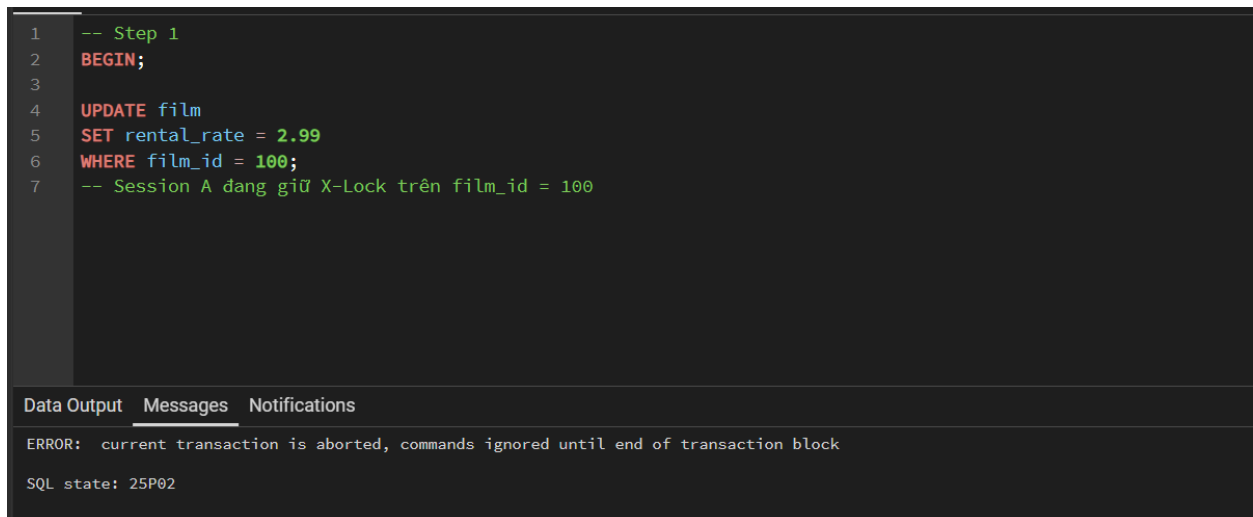


The screenshot shows a PostgreSQL query editor window titled 'dvdrental/postgres@PostgreSQL 18'. The query is as follows:

```
1  --Step 2
2  BEGIN;
3  UPDATE film SET rental_rate = 5.99
4  WHERE film_id = 1;
5  --BLOCKS IMMEDIATELY!
6  --Waits for Session A
7
8  --Step 4 (Observing)
9  --Executes automatically after Acommits.
10 COMMIT;
```

The 'Messages' tab is selected, showing the output: 'COMMIT' and 'Query returned successfully in 72 msec.' A green status bar at the bottom right confirms: '✓ Query returned successfully in 72 msec. ✕'. The bottom status bar indicates 'Total rows: Query complete 00:00:00.072' and 'GBLE | Ln 10, Col 1'.

Lab 3 :



The screenshot shows a PostgreSQL query editor window with the following query:

```
1  -- Step 1
2  BEGIN;
3
4  UPDATE film
5  SET rental_rate = 2.99
6  WHERE film_id = 100;
7  -- Session A đang giữ X-Lock trên film_id = 100
```

The 'Messages' tab is selected, displaying an error: 'ERROR: current transaction is aborted, commands ignored until end of transaction block' and 'SQL state: 25P02'.

→ Chưa COMMIT, giữ nguyên tương tự với cả 2 session A,B

Query History

```

1  --Step 2
2  BEGIN;
3  UPDATE film SET rental_rate = 2.99
4  WHERE film_id = 200;
5  --Step 4
6  UPDATE film SET rental_rate = 3.99
7  WHERE film_id = 100;
8  --BOOM! ERROR: deadlock detected

```

Data Output Messages Notifications

ERROR: deadlock detected
Process 24044 waits for ShareLock on transaction 995; blocked by process 25436.
Process 25436 waits for ShareLock on transaction 996; blocked by process 24044.

SQL state: 40P01
Detail: Process 24044 waits for ShareLock on transaction 995; blocked by process 25436.
Process 25436 waits for ShareLock on transaction 996; blocked by process 24044.
Hint: See server log for query details.
Context: while locking tuple (83,8) in relation "film"

✓ Query returned successfully in 39 secs 526 msec.

Total rows: Query complete 00:00:01.101 CRLF Ln 6, Col 1

→ Lúc này Deadlock xảy ra (A chờ B, B chờ A).

Lab 2 :

Query History

```

1  BEGIN;
2  SELECT rental_rate FROM film
3  WHERE film_id = 10

```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	rental_rate numeric (4,2)
1	4.99

→ Chưa COMMIT, giữ nguyên transaction đang mở.

```
1 BEGIN ;
2 UPDATE film SET rental_rate = 0.99
3 WHERE film_id = 10;
4 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 72 msec.

```
1 SELECT rental_rate FROM film
2 WHERE film_id = 10;
3 --NEWRESULT: 0.99!
4 COMMIT;
```

Data Output Messages Notifications

WARNING: there is no transaction in progress

COMMIT

Query returned successfully in 74 msec.

-- Kết quả mới: 0.99 !

Lab 4 :

Query Query History

```
1
2 ROLLBACK;
3 BEGIN;
4 SELECT * FROM inventory
5 WHERE inventory_id = 10 FOR UPDATE ;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	Inventory_Id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

→ Chưa COMMIT, giữ nguyên transaction.

Query Query History

```
1 ROLLBACK ;
2 BEGIN ;
3 SELECT * FROM inventory
4 WHERE inventory_id = 10 FOR UPDATE ;
5
```

Data Output Messages Notifications

Total rows: Waiting for the query to complete... 00:00:02.662 CRLF Ln 1, Col 12

→ Session B sẽ treo (không trả kết quả).

```
6
7 INSERT INTO rental (rental_date, inventory_id, customer_id, staff_id)
8 VALUES (NOW(), 10, 1, 1);
9
10 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 177 msec.

✓ Successfully run. Total query runtime: 2 min 3 secs. 1 rows affected. ✕

✓ Query returned successfully in 177 msec. ✕

-- Giải phóng lock sau khi đã thuê thành công

Sau khi Session A COMMIT, Session B sẽ được chạy tiếp:

```
1 ROLLBACK ;
2 BEGIN ;
3 SELECT * FROM inventory
4 WHERE inventory_id = 10 FOR UPDATE ;
5
6 ROLLBACK;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 104 msec.